

Analysis First-Year Exam, May 2024, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable, let g be a real-valued function on $[a, b]$, and let $D = \{t \in [a, b] : f(t) \neq g(t)\}$.
 - (i) Show by example that if D is countably infinite then g can fail to be Riemann integrable.
 - (ii) Prove that if D is finite then g must be Riemann integrable.

2. Prove Dini's theorem that, if the sequence $(f_n)_{n=0}^\infty$ of continuous real-valued functions on a compact space decreases pointwise to zero, then the convergence is uniform. Show by example that uniform convergence can fail if decreases pointwise is replaced by converges pointwise.

3. State the Weierstrass approximation theorem. Determine exactly which continuous functions $f : [-1, 1] \rightarrow \mathbb{R}$ satisfy the requirement that for each strictly positive integer n ,

$$\int_{-1}^1 f(t)t^n dt = 0.$$

4. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and let $(A_n)_{n=1}^\infty$ be a sequence in $\mathcal{A}$. For each positive integer N , let B_N be the set containing exactly those $\omega \in \Omega$ that lie in A_n for at least N values of n . By considering the series $\sum_{n=1}^\infty \mathbf{1}_{A_n}$ of indicator functions, prove that

$$N\mu(B_N) \leq \sum_{n=1}^\infty \mu(A_n).$$

5. Let the non-negative function $f : \mathbb{R} \rightarrow \mathbb{R}$ be integrable with respect to Lebesgue measure λ and for each real t define

$$F(t) = \int_0^t f d\lambda.$$

Prove that the function F is continuous. Must F be uniformly continuous? Explain.